



Thermal power plant upgrade via a rotating detonation combustor and retrofitted turbine with optimized endwalls

Zhe Liu, James Braun*, Guillermo Paniagua

Purdue University, 500 Allison Rd, West Lafayette, IN 47907, United States

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ABSTRACT

In the past decade, pressure gain combustion research has promised over 10 percentage-points of increase in power plant thermal efficiency. Alas, to realize such potential gain, one must effectively couple the turbine with the detonation combustor, whose exhaust conditions differ substantially from current state of the art gas turbines. This paper presents a modeling approach that enables a superior thermodynamic cycle with a rotating detonation combustor and a retrofitted gas turbine by including a diffuser downstream of the combustor and by contouring the turbine endwall, while preserving the airfoil geometry. We propose a multi-step optimization strategy, parametrizing the endwall geometry with a few control points, without altering the airfoil geometry, and with the stage turbine efficiency as objective function. In a first step the turbine performance is assessed with steady inlet conditions by solving the steady three-dimensional Reynolds-Averaged Navier-Stokes equations. In a second step, the inlet conditions are unsteady, as predicted from a detonation combustor and a diffuser, and three-dimensional full unsteady simulations are performed with an unsteady Reynolds-Averaged Navier-Stokes solver. By altering the vane endwall, the steady optimization yielded an efficiency increase of 12% relative to the baseline, while the unsteady optimization resulted in 21% increase compared to the datum turbine. Finally, a full engine analysis demonstrated the superiority of pressure gain combustion which included a realistic thermodynamic cycle of the combustor, the diffuser and the optimized turbine components.

1. Introduction

The U.S. Energy Information Administration calculated that around 10.59 trillion cubic feet of natural gas was used for power generation in 2018 and accounts for the largest share of greenhouse gas emissions (Data source: U.S. EPA, 2016). Currently, efficiencies for combined cycle gas-fired plants lie around 56–64% and are based on deflagrative combustion. A potential candidate for fuel-reduction lies in replacing the conventional combustor with a pressure gain combustor which offers less fuel consumption compared to the conventional deflagration-based engines in which combustion occurs through an isobaric process [1]. Two of the most promising combustion architectures to realize pressure gain are the pulse detonation combustors (PDC), in which a tube is cyclically filled with fuel-oxidizer mixture and is then detonated, and rotating detonation combustors (RDC) in which the combustion mixture is injected axially and the ignition is self-sustained by a rotating detonation wave. Although performance benefits with pressure gain combustion with idealized components have been reported since early 2000 and demonstrated values over 20 percentage-points of efficiency increase [2], the characterization of the turbine interaction with pressure gain combustors to measure the actual performance of the turbine has only

started recently. Alternative architectures rely on organic Rankine cycles coupled with radial turbo-expanders [3–5].

Xisto et al. [6] numerically characterized a coupled axial turbine-pulse detonation tube system. The inlet of the turbine in this configuration was exposed to transonic flow at the occurrence of the detonation wave and decreased to low subsonic values downstream. Results indicated that fluctuations of the rotor incidence angle were the main contributors to the losses. Additionally, the turbine efficiency was highest during the purge conditions and suggestive of improved performance with ad-hoc turbines. Anand et al. [7] experimentally assessed a system consisting of an axial turbine and six pulse detonation tubes. The thermal efficiency of the system showed an increase compared to the Brayton cycle but the turbine efficiency was highly reduced due to presumed flow angle variations across the rotor passages. The practical integration of PDCs into gas turbines is however constrained by the cyclic refill and detonation ignition and lower power density [8] compared to rotating detonation combustors (RDCs).

These RDCs have been studied with increased modeling complexity, with two-dimensional Euler [9] to three-dimensional premixed unsteady Reynolds averaged Navier Stokes solvers [10] to explain the complex flow physics. Meng et al. [11] proposed unsteady RANS simulations to

* Corresponding author.

E-mail address: jamesbraun91@gmail.com (J. Braun).

Nomenclature

A	Oscillating amplitude of stator inlet total pressure fluctuations [-]
C	Chord length [mm]
C_p	Constant pressure specific heat [J/(kg.K)]
f	Oscillating frequency of stator inlet total pressure fluctuations [kHz]
$\bar{f}$	Reduced frequency [-]
g	Pitch [mm]
F	Thrust [N]
$\dot{m}$	Mass flow rate [kg/s]
M	Absolute Mach number [-]
η_{thermo}	Turbine efficiency [-]
η_c	Compressor efficiency [-]
P_0	Total pressure [bar]
Q_l	Lower heating value [J/kg]
π	Compression ratio [-]
R	Radius [mm]
Re	Reynolds number [-]
S	Area [mm ²]
t	time [s]
$t_{periodic}$	time to form a periodic cycle [s]
T_0	Total temperature [K]
T_s	Static temperature [K]
V	Flow velocity in the absolute frame of reference [m/s]
x	Design parameter [-]
y^+	Non-dimensional wall distance
λ	Normalized torque residual [-]

Abbreviation

CADO	Computer Aided Design and Optimization
DoE	Design of Experiments
GCI	Grid convergence index
RANS	Reynolds-Averaged Navier-Stokes
RDC	Rotating detonation combustor
RPM	Revolutions per minute
SST	Shear stress transport
URANS	Unsteady Reynolds-Averaged Navier-Stokes
T-MATS	The Toolbox for the Modeling and Analysis of Thermodynamic Systems

Subscripts

ax	axial direction
c	chord
f	fuel
R	Relative frame of reference
h	Hub
is	isentropic
m	mean value
t	Tip
0	Engine inlet
1	Stator inlet
2	Stator outlet
3	Rotor outlet
5	Engine outlet

study non-premixed rotating detonation combustors. In addition to a primary rotating shock, several other RDC modes exist such as counter rotating shocks as highlighted by Bohon et al. [12]. Other additional shock-reflections could occur, as explained by Zhou et al. [13] and depend on chamber geometry such as chamber width. Kawasaki et al. [14] found a minimal inner diameter for efficient thrust generation. Hence, a set of optimal parameters that define the geometry and inlet conditions could exist. To identify those parameters, reduced models for

the RDC are proposed by Sousa et al. [15] and Fievisohn et al. [16] based on one-dimensional reaction models and the method of characteristics. Sousa et al. [15] identified the optimal radius-over-detonation-height that yields minimal entropy generation across the RDC. Kaemming et al. [17] proposed a reduced-order model including several loss mechanisms such as deflagrative burning. Mizener et al. [18] performed a parametric study of the RDC to include waveriders for high-speed applications. Most RDC research involves annular architectures. However, Najagami et al. [19] discussed other geometries such as disk-shaped rotating detonation combustors or hollow rotating detonation combustors [20]. The exhaust of rotating detonation combustors can be altered via various nozzles and diffusers as demonstrated by Goto et al. [21]. To define the outlet characteristics of those combustors and nozzles, Braun et al. [22] calculated that Mach number at the outlet of RDCs with a straight nozzle section periodically fluctuates around the transonic region. This poses a serious issue for turbine designers, as typically, inlet Mach numbers to the turbine are around 0.1. Direct coupling of a subsonic turbine with an RDC will result in an unstarted operation because of the high inlet Mach numbers, which leads to an unsteady shock wave that moves upstream and causes significant total pressure loss as demonstrated in [23].

Several solutions can be proposed to overcome these unstarting phenomena and to efficiently harness the power from the transonic flow out of RDCs, as depicted in Fig. 1. A first option is to mount a supersonic axial turbine, characterized by slender airfoil shapes and low turning. In this case, a nozzle [22] is attached upstream of the turbine to accelerate the flow to Mach 2 and to reduce the fluctuations. Paniagua et al. [23] proposed a new class of high-speed supersonic axial turbines and Sousa and Paniagua [24] implemented a method of characteristics solver to optimize these supersonic internal passages with minimum entropy. Liu et al. [25] characterized the performance of a supersonic axial turbine exposed to periodic oblique shocks and results revealed that the leading edge shock loss was the main contributor to the loss mechanism and greatly reduce the efficiency. To estimate fuel savings, Sousa et al. [1] demonstrated via a detonation-based engine model equipped with a supersonic turbine that the thermal efficiency could be increased by up to 8 percentage-points at low pressure ratios compared to the deflagration-based engine.

A second option is to mount a high-efficient subsonic axial turbine. Therefore, a diffuser is required upstream of the turbine to decelerate the flow from supersonic to the high subsonic regime. To operate axial turbines at high subsonic inflow (an inlet Mach number of 0.6), Liu et al. [26] suggested to open the stator endwalls by 70 percentage-points and obtained a turbine efficiency of 80% while a constant endwall design showed a deficient performance of less than 50% due to the failure of respecting the isentropic limit. However, the increased area opening introduced flow separation close to the stator endwalls. Unsteady recirculation bubbles interacted with the stator secondary flow under pulsating environments and consequently penalized the turbine aerodynamic performance. Other studies about axial turbines exposed to inlet fluctuations [27,28] highlight a wide span of turbine efficiency depending on the frequency and amplitude of inlet fluctuations, from 61% for high amplitudes [27] to 83% for low amplitudes [28]. In addition, Fernelius et al. [28] identified the fluctuating amplitude to be the main cause of the efficiency decrease. Naples et al. [29] tested a turbine configuration downstream of a rotating detonation combustor with a dilution over RDC air flow in excess of 350% and found efficiencies below 80%, similar to the conventional deflagration combustor. The values reported in these studies are well below state-of-the-art turbine design, in which turbines are designed with efficiencies in excess of 90% [30,31].

Hence, in this paper, a multi-step optimization strategy [32] is proposed to boost the efficiency of turbines for rotating detonation combustors by retrofitting the endwall of a subsonic axial turbine to achieve the promised increase in cycle efficiency. The stator endwall is parametrized while turbine airfoils are unchanged. First, a baseline profile with end-wall diffusion [26] is optimized with steady-inlet Reynold Averaged

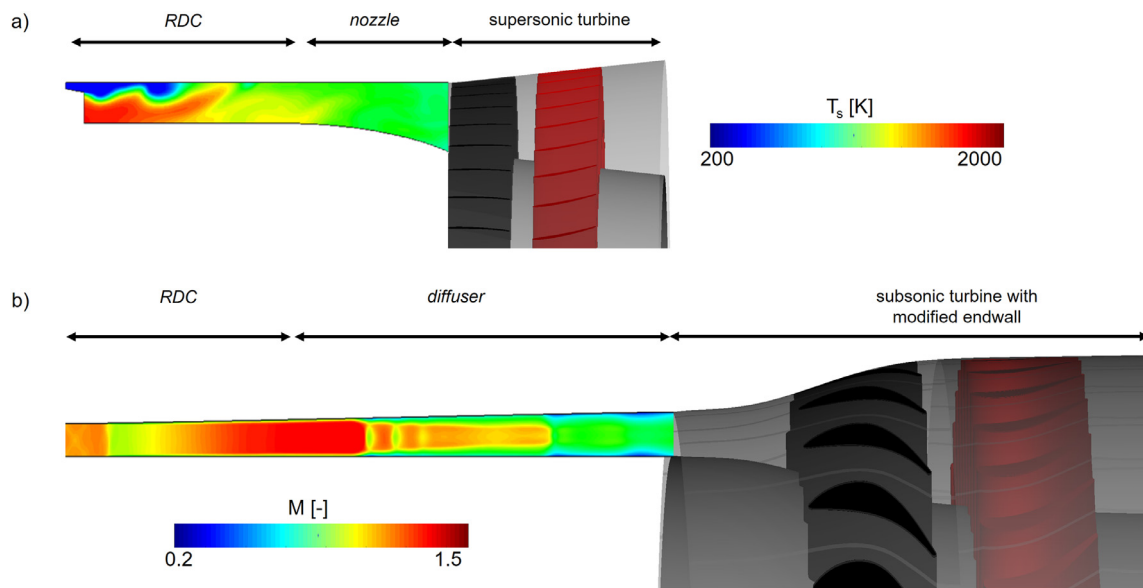


Fig. 1. Fluid expansion systems downstream a rotating detonation combustor: a) supersonic turbine + nozzle and b) subsonic turbine + diffuser.

Navier Stokes (RANS) simulations. The objectives are to enhance turbine efficiency and specific work. Second, the selected optimized profiles for steady inlet are further improved using unsteady RANS simulations with a fluctuating inlet. Afterwards, full unsteady simulations are performed on selected optimized turbines to retrieve the stage efficiency, power and damping. Finally, an engine analysis that includes a realistic detonation-diffuser-subsonic turbine model is used to predict the enhancement of the retrofitted turbine for high Mach number inflows and pressure gain combustor by comparing this to the baseline design and to the traditional Brayton cycle engine.

2. Methodology of turbine optimization

2.1. Diffuser outlet conditions/turbine inlet conditions

Prior research [26] demonstrated that inlet Mach numbers of 0.3 yielded efficiencies up to 90% at the expense of lengthy diffusers upstream of the turbine, but efficiencies below 80% for higher inlet Mach numbers than 0.6 with shorter diffusers. In the current optimization, the stator inlet Mach number is 0.6, as delivered by a short diffuser. In the relative frame of reference, the rotor inlet relative Mach number is 0.45, to enable a turning of 100 deg. The span increases from inlet to outlet about 10% to ensure the rotor transonic operation at a turbine total-to-static pressure ratio of 3. Throughout this optimization the rotor passage is unaltered.

2.2. Turbine optimization strategy

The multi-step optimization strategy [32] is outlined in Fig. 2: The steady optimization starts with a turbine parametrization via an in-house code [33]. In this case, seven design parameters were selected for the stator and rotor endwalls. The blade and endwall profiles are parametrized with Bezier curves and 3D blades are obtained by stacking the 2D profile at certain radius [32]. Consequently, the turbine geometry is meshed via Autogrid of Numeca and the RANS equations are solved. The objective functions (thermodynamic efficiency and specific work) are extracted from the post-processing tools and are fed to a differential evolution optimizer, CADO [34]. In the second step, the three most influential design parameters of the end wall profiles from the steady optimization are selected for the next step unsteady optimization. During the unsteady optimization, the parametrization and meshing follow the

same methodology as the steady routine. The URANS equations were solved with the large turbine inlet fluctuations as inlet boundary conditions, delivered by the short diffuser. The vane exit conditions are pitch-wise averaged at the mixing plane, and subsequently transposed to the rotor inlet [32]. As final verification, full unsteady simulations of the turbine stage were carried out on a few selected optimized configurations by modeling two stator passages and three rotor passage without tangential averaging at the stator-rotor interface to resolve the stator-rotor interaction. Two amplitudes of the unsteadiness were modeled and are indicative of the turbine passage sensitivity to unsteadiness. The main driver to loss creation is flow separation occurring across the vane passage endwalls [26]. Hence, the present optimization strategy focuses on stator endwall contouring. For the optimal turbine configuration, the vane kinetic energy losses ($\frac{v_{2, is}^2}{v_2^2} - 1$) are reduced three times more than the rotor kinetic losses. An alternative optimization strategy that could potentially provide higher efficiency should involve the control of the rear vane diffusion rate along the entire span vane, which in turn would require the optimization of the rotor shape to accommodate the different radial distribution of rotor inlet flow angle.

2.3. Selection of the turbine vane geometry and endwall parameterization

The turbine geometry was obtained by scaling a high-pressure transonic turbine, with a pitch-to-chord ratio of 0.75 [35]. Fig. 3 illustrates the turbine channel parametrization [32] for the steady optimization. The vanes and rotor blade profiles were unaltered during the optimization and the stage had a degree of reaction of 0.3 [34]. This airfoil design was selected for its adequate performance at on and off-design condition and was validated in various experimental and numerical studies [30,31,34,36] in contrast to impulse blades which trade turbine efficiency for work extraction and perform poorly at off-design conditions. Table 1 provides the geometric parameters of the turbine airfoils and the entire stage encompasses a total-to-static pressure ratio of 3.

Four free control points (x_1, x_2, x_3 , and x_4) defined the contour of the stator endwall and were free to move in the radial direction with a certain degree of variability. Three free control points (x_5, x_6 , and x_7) were assigned to contour the rotor hub. The fixed points were constrained in both radial and axial direction to allow for a Mach 0.6 flow at the inlet of the turbine. The endwalls were contoured with a Bezier curve to ensure a smooth endwall design and axisymmetry was assumed. The steady optimization was initialized with a Design of Experiment through

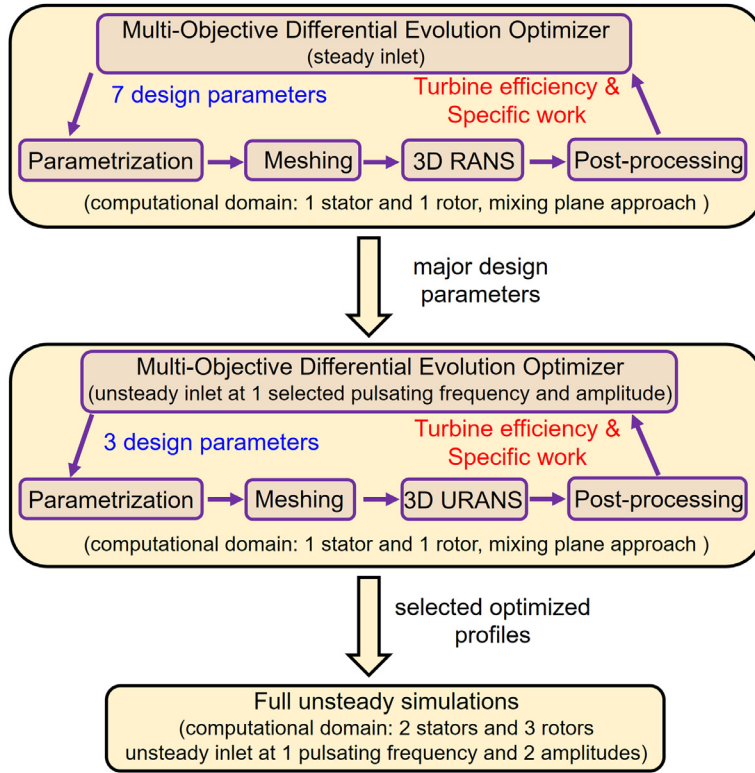


Fig. 2. Multi-step optimization strategy: step 1 – steady optimization via RANS simulations; step 2 – unsteady optimization via URANS simulations with critical design parameters identified from step 1; step 3–full unsteady evaluation of selected optimized profiles.

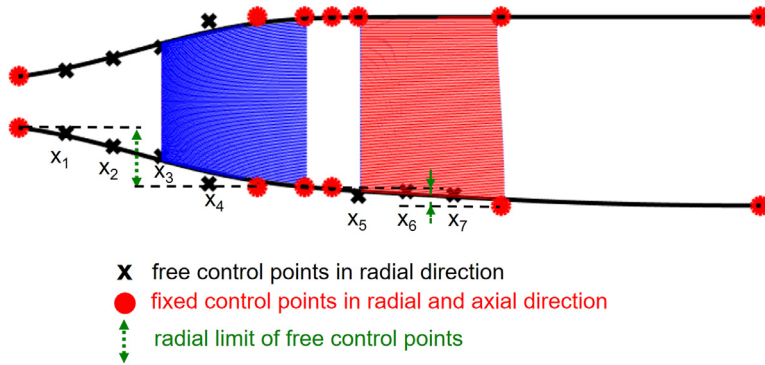


Fig. 3. Turbine endwall parametrization with seven design parameters for the steady optimization.

Table 1

Mid-span geometric parameters of the investigated transonic turbine stage.

	stator	rotor
vane/blade count	43	64
C_{ax} [mm]	55.73	51.08
h/C_{ax} [-]	1.23	1.36
R_t/R_h [-]	1.15	1.16
Stagger [deg]	52	32
RPM [-]	-	10421
Re_c [-]	$2.23 \cdot 10^6$	-

a fractional factorial method [37] to generate 128 individuals. The individuals per population of the multi-objective optimizer, based on the NSGA-II algorithm [38], were set to 40.

2.4. Description of the computational domain and solver

CFD++ from Metacomp Technologies [39] was chosen to solve the 3D URANS equations. The turbulence closure was provided by the $k-\omega$

SST model and the first layer thickness, y^+ , was lower than 1 to resolve the viscous sublayer. The solver was previously validated for a transonic stator case to demonstrate the prediction of the shock location [24] and a Mach 5 ramp case to predict the shear stress [25]. Braun et al [40] used CFD++ with the $k-\omega$ -SST turbulence model for a Mach 2 bladeless turbine and could accurately track the shocks and separation onset. However, due to the limitations of URANS, the separation region was mispredicted by up to 13%. The original cross-section of the channel along the stator and rotor row is visualized in Fig. 4a and across the rotor, the channel is slightly opened. Fig. 4b shows a profile in which the inlet area is diffused by 70 percentage-points and allows to ingest inlet Mach numbers of around 0.6 [26]. From a steady analysis, the thermodynamic efficiency was below 80% [26] and this profile (Fig. 4b) is defined as the baseline profile. Fig. 5 illustrates the computational domain of the baseline turbine with a detailed mesh topology on the stator trailing edge and rotor leading edge. A structured grid was generated with Autogrid5 from NUMECA International, and contained around 7 million cells. A mixing plane at the stator-rotor interface performs a tangential averaging on the fluctuations from the upstream stator and then conveys them to the downstream rotor [32]. The mesh indepen-

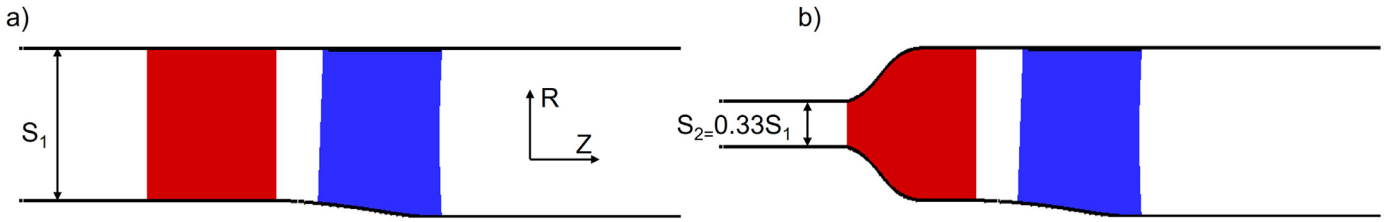


Fig. 4. Turbine enwall variation with a) constant stator endwall and b) modified stator endwall ('baseline').

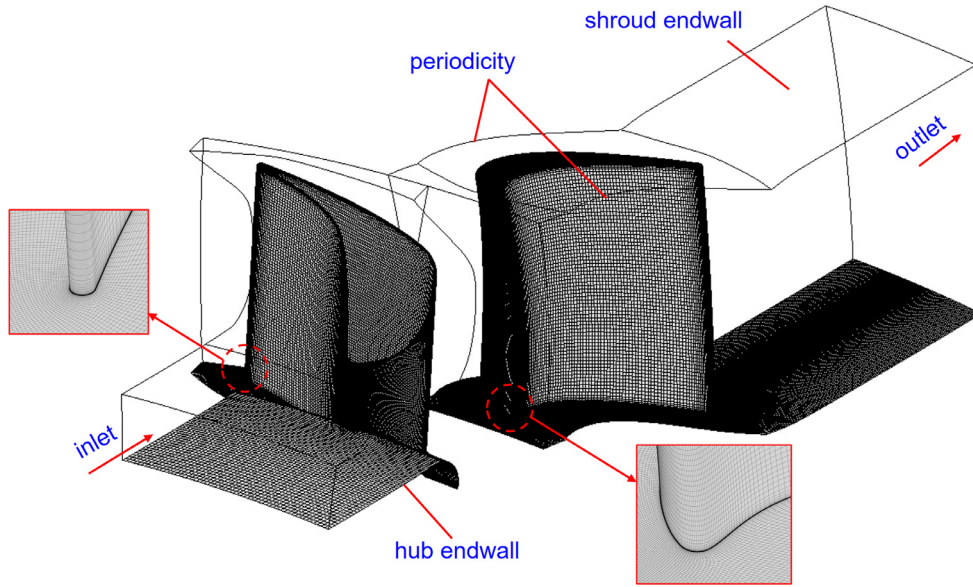


Fig. 5. Computational domain of the baseline turbine stage for an inlet Mach number of 0.6, with snapshots of the stator trailing edge and the rotor leading edge mesh topology.

dence study was reported in [32]. For the steady optimization, constant total pressure and total temperature were prescribed at the stator inlet, while constant static pressure was imposed at the rotor outlet.

2.5. Description of the diffuser outlet boundary conditions

To generalize the unsteady results, the reduced frequency [26,32] is used which compares the inlet excitation frequency to the frequency at which convective properties propagate across the stator passage. The axial velocity is instantaneously mass-flow-averaged over all grid points (i) at the stator inlet (Eq. (1)) and afterwards, a cycle-mass-averaged evaluation is performed (Eq. (2)) over one pulsating period. Unsteady convergence was ensured with the method described by Clover and Clark [41]. Finally, the reduced frequency (Eq. (3)) was evaluated via the ratio of the excitation frequency and the frequency at which convective characteristics traverse. The Purdue Rice cluster includes 580 HP nodes and each node has 20 cores Intel Xeon-E5 processors with 64 GB of memory [42]. The convergence time is 10 hours on one node for a RANS stage simulation, 48 hours on one node for a URANS simulation with the mixing plane approach, and around 170 hours on three nodes for a full unsteady simulation.

$$\bar{V}_{ax,in}(t) = \frac{\sum_{i=1}^n V_{ax,in}(t) \rho_i |\bar{V}_{ax,i} \cdot \bar{S}_i|}{\sum_{i=1}^n \rho_i |\bar{V}_{ax,i} \cdot \bar{S}_i|} \quad (1)$$

$$\bar{V}_{ax,in} = \frac{\int_0^t m(t) \bar{V}_{ax,in}(t) dt}{\int_0^t m(t) dt} \quad (2)$$

$$\bar{f} = \frac{f}{V_{1,ax}/C_{ax,S}} \quad (3)$$

The mean-to-peak amplitude of the inlet fluctuations (the difference between the maximum and mean value during one period) is computed

by Eq. (4) and is a quantitative measurement of the strength of the pulsations [32].

$$A = \frac{Max - Mean}{Mean} \quad (4)$$

The boundary conditions were provided via Eq. (5). Typical frequencies for RDCs are in the kHz frequency range for h_2 -air [43], natural gas-hydrogen fuel blends [44] for syngas-air mixtures [45]. For the baseline configuration [26], the turbine had the lowest efficiency (72%) when the stator inlet was exposed to pulsations at the intermediate reduced frequencies ($0.1 < f < 1$) and highest investigated amplitudes. Within this frequency band, the speed of inlet excitation traveling across the stator has the same order of magnitude as the propagation of convective properties and this operation was most sensitive to efficiency drop [26], hence, this excitation level was selected as boundary condition for the unsteady optimization.

$$P_{01} = P_{0m} + \sin(2\pi f) A P_{0m} \quad (5)$$

3. Results

3.1. Turbine steady optimization

During the steady optimization, six populations were generated with a total of 368 geometries. Fig. 6 depicts the 368 geometries that were evaluated, with a color-level related to the specific work. Turbine efficiency and specific work are evaluated via Eqs. (6) and (7). Profiles with high inlet Mach numbers in the upper right area also show high efficiency and specific work and optimal inlet Mach numbers range between 0.63 and 0.65. Optimized endwall geometries have around 13% enhancement in efficiency and specific work. In addition, the position of the control points from the superior endwall designs suggest that the free control point x_1 of the stator endwalls (Fig. 3) should be radially

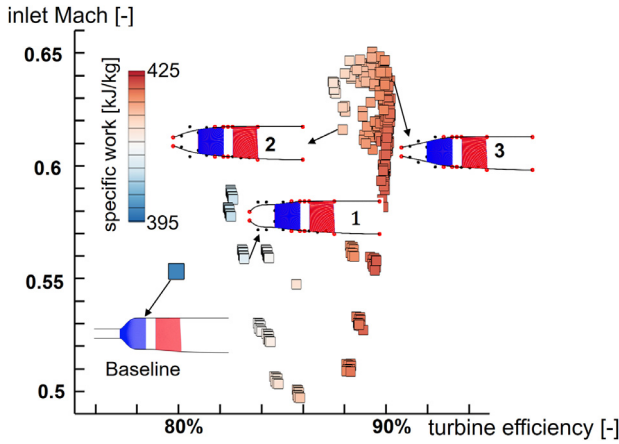


Fig. 6. Results of the steady endwall optimization plotted by turbine efficiency and inlet Mach number.

close to the first fixed control point and the rotor hub endwall control points (points x_5 , x_6 , and x_7 in Fig. 3) have a minor effect on the turbine performance. Hence, three major design parameters (x_2 , x_3 , and x_4) are selected for the unsteady optimization.

$$\eta_{thermo} = \frac{T_{01} - T_{03}}{T_{01} - T_{03, is}} \quad (6)$$

$$w = C_p (T_{01} - T_{03}) \quad (7)$$

Where the turbine outlet isentropic total temperature $T_{03, is}$ is assessed via Eq. (8):

$$T_{03, is} = T_{01} \left(\frac{P_{03}}{P_{01}} \right)^{\frac{\gamma-1}{\gamma}} \quad (8)$$

Separation due to endwall diffusion causes total pressure loss and negatively impacts the efficiency and work potential of the turbine stage. This effect is amplified for increased diffusion across the stator [26]. Therefore, the amount of separation at a cross-section plane within the stator passage was assessed and is defined in Eq. (9) as the amount of area that contained separated flow [32]. Separated flow area was defined as a mesh cell in which the axial velocity is less than 0.01 m/s.

$$relative \ size \ of \ separated \ flow = \frac{S_{(V_{ax} < 0.01)}}{S} \quad (9)$$

Fig. 7a describes a pitch-wise cut within the baseline stator passage contoured by the axial velocity. A significant separation occurs for the baseline stator with a relative size of separated flow of 19.5% while the steady-optimized configuration shows a value of only 0.2%. Fig. 7b displays the separation quantification. The upper right zone, featured by profiles with high efficiency and specific work, is also characterized by small separated flow areas.

3.2. Turbine unsteady optimization

Seven populations were created which resulted in 37 endwall geometries. To evaluate the objective functions for the unsteady cases, all parameters of Eq. (8) were first instantaneously mass-flow-averaged and then cycle-mass-averaged over a periodic cycle to derive the turbine outlet isentropic total temperature. Afterwards, the turbine outlet isentropic total temperature (Eq. (8)) was evaluated and used to determine the thermodynamic efficiency (Eq. (6)), together with other cycle-mass-averaged of mass-flow-averaged quantities. Damping is used to quantify the residual fluctuations at the outlet of the first turbine stage and was

assessed with Eq. (10) [32].

$$Damping_{p_0} = \frac{A_1 - A_3}{A_1} \quad (10)$$

Fig. 8a displays the damping and efficiency of the same 37 turbine configurations, and the color level designates the specific work. Profiles with highest efficiency and damping appear in the upper right area, with a maximum damping of around 85%. Stage damping of profile 2 is more than 5 percentage-points higher than that of profile 3 at the expense of the turbine efficiency of profile 2 which is 1.4 percentage-points lower. For further analysis, profile 2 was selected. Compared to the baseline profile [26], the turbine efficiency increased by approximately 21.4% for the optimized endwall geometries. The specific work from the optimized configurations was augmented by more than 22%. From the baseline to the optimized profiles, damping of the total pressure fluctuations increased by up to 30 percentage-points. Fig. 8b plots the total pressure signature extracted from the stator inlet and rotor outlet for both baseline and unsteady-optimized turbine (profile 2, Fig. 8a) for two pulsating periods, where the stage damping across profile 2 was 83% and was 29 percentage-points higher than the baseline case.

For the baseline profile, the endwall opening only occurs across the stator passage. As a result, flow experiences diffusion at the beginning and is then accelerated until the stator trailing edge. However, for both the steady and unsteady optimized profiles, flow is diffused upstream of the stator in such a way that flow turning is enhanced and full acceleration is achieved in the stator passage. Fig. 9a–c display the channel for the baseline, the steady-optimized (profile 3, Fig. 6), and unsteady-optimized (profile 2, Fig. 8a) stator, and the stator throat-to-local area ratio for each case is plotted in Fig. 9d. For the baseline stator, the local cross-section area keeps increasing from the leading edge till the mid-axial chord during which diffusion is dominant and flow separation occurs whereas the channel keeps converging for both optimized configurations from the stator leading edge till the trailing edge. In addition, at steady state, the Mach number in front of the stator leading edge is 0.64 for the baseline, 0.3 for the steady-optimized, and 0.35 for the unsteady-optimized profile. This indicates that the unsteady-optimized endwall is diffused to a lesser extent upstream of the stator leading edge, which helps to suppress the growth of the unsteady recirculation bubbles for the pulsating inlet condition compared to the steady-optimized endwall geometry. Under such large inlet fluctuating conditions, the unsteady-optimized turbine significantly outperforms the baseline, the vane time-averaged of the mass-flow-averaged total pressure loss was reduced by 6.7 percentage-points. Two steady simulations were performed in which the turbine was subjected to the highest and lowest pressure ratio of the pulsating period. An efficiency variation of around 0.5 percentage-point was retrieved for the unsteady optimized turbine while this increased to 8.4 percentage-points for the steady optimized profile and 8 percentage-points for the baseline turbine, suggesting that a multi-point optimization could provide a cost-effect intermediate solution by performing a steady evaluation for each steady-optimized geometry at three different inlet total pressures (lowest, mass-flow averaged mean and highest). However, the unsteady-optimized turbine (profile 2, Fig. 8a), when operating at a large amplitude fluctuation, displays an efficiency decrease of 7.7 percentage-points due to the unsteadiness compared to the steady state, which cannot be accounted for by a series of steady evaluations.

3.3. Full unsteady assessment of optimized endwall geometries

Full unsteady simulations with two stator passages and three rotor passages are performed at a reduced frequency of 0.23 for two peak-to-mean amplitudes on selected optimized turbines without tangential averaging at the stator-rotor interface to resolve the stator-rotor interaction. The entire computational domain contains two stator passages and three rotor passages. Consequently, the computational burden increased to 170 hours on three Intel Xeon-E5 nodes compared to 48 hours for the mixing plane approach with only one node. Profile 3 from Fig. 6 was

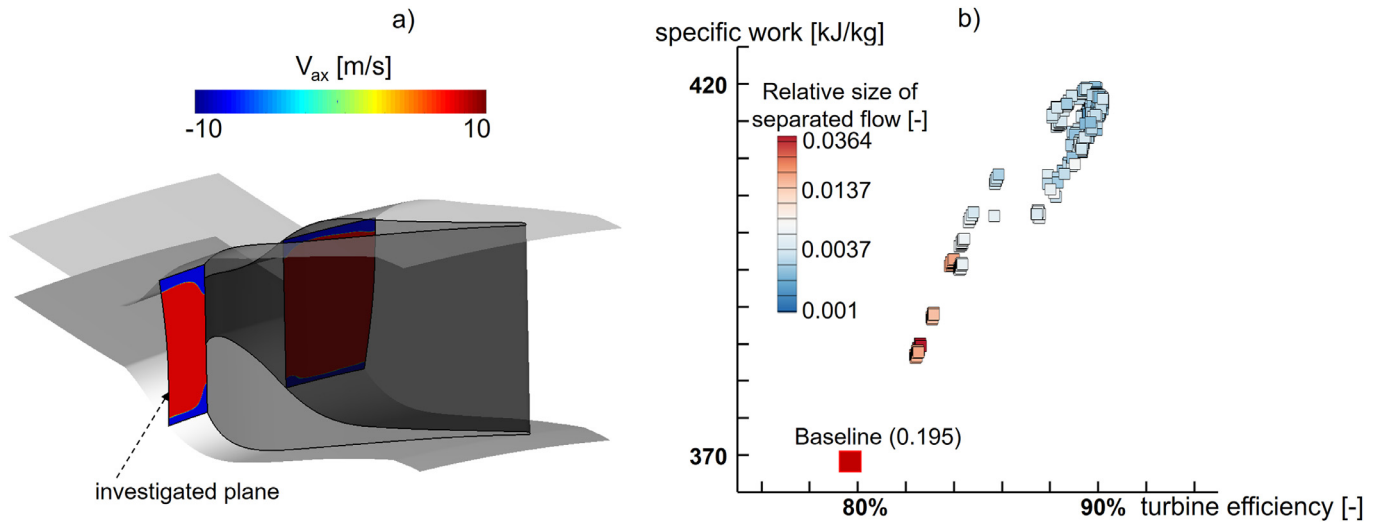


Fig. 7. a) Axial velocity contour at a certain pitch-wise cut within the stator passage of the baseline endwall design. b) Relative size of separated flow (log scale) from the steady endwall optimization.

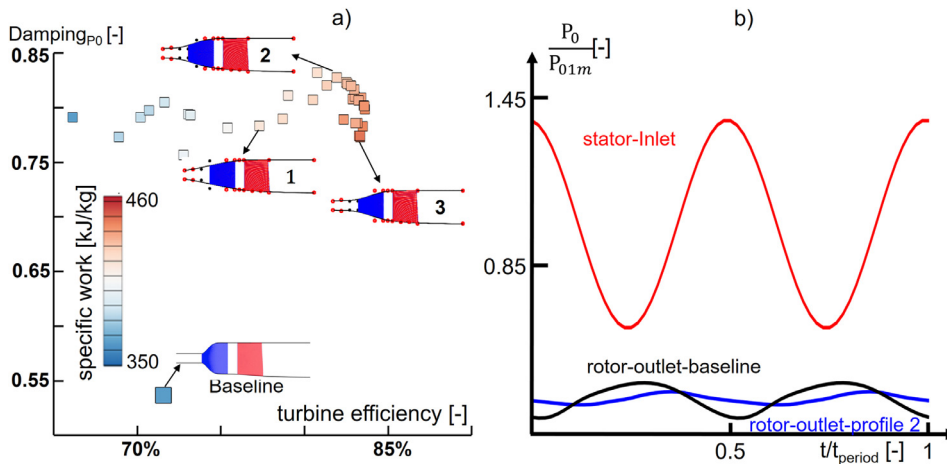


Fig. 8. a) Results from the unsteady endwall optimization colored by specific work. b) Total pressure traces at the stator inlet and rotor outlet for the baseline and optimized endwall profile.

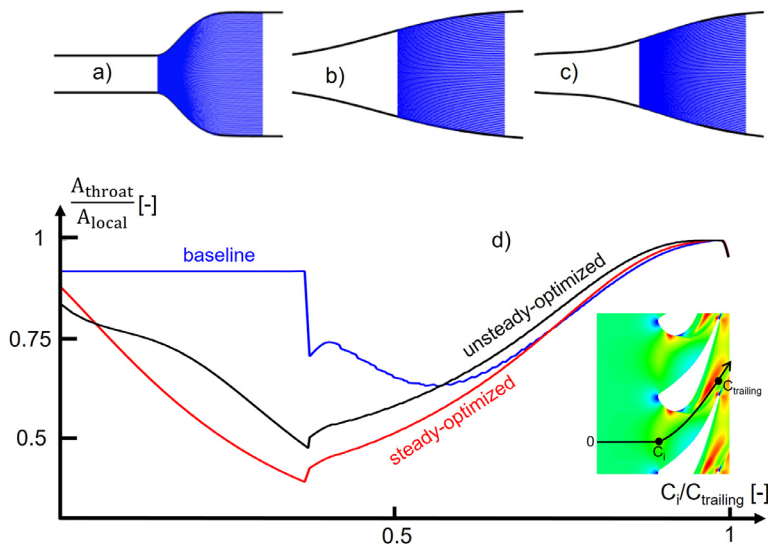


Fig. 9. Endwall variation of a) baseline stator, b) steady optimized profile (profile 3 in Fig. 6), and c) unsteady optimized profile (profile 2 in Fig. 8a). d) Throat-to-local area ratio in function of the normalized stator chord length.

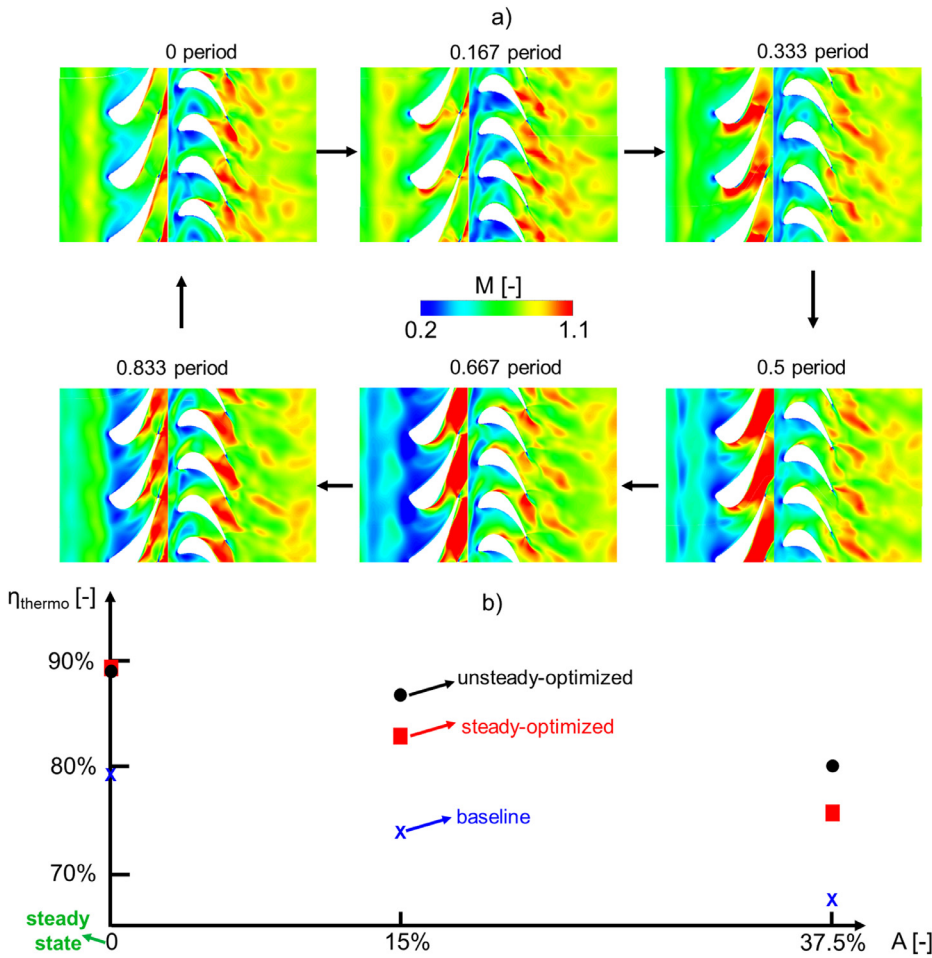


Fig. 10. a) Flow field from a full unsteady turbine simulation (profile 3, Fig. 6) at $A=37.5\%$ and $\bar{f}=0.23$. b) Turbine efficiency of the baseline and selected optimized configurations in function of the amplitude of the inlet total pressure oscillations.

Table 2
Total pressure loss of the stator and rotor of the baseline and unsteady-optimized profile.

	1- P_{02}/P_{01}		1- P_{03R}/P_{02R}	
	baseline	unsteady-optimized	baseline	unsteady-optimized
steady state	11.6%	4.5%	7.6%	5.2%
A=15%	12.7%	5.2%	6.7%	5.6%
A=37.5%	15.5%	8.8%	8.1%	7.8%

selected for the steady-optimized profile while profile 2 from Fig. 8a was chosen for the unsteady-optimized profile. In all sections, the degree of reaction was evaluated using Eq. (11). For the unsteady calculation, the static and total enthalpy were first massflow averaged at each plane and then time averaged. Under the large amplitude fluctuating condition ($A=37.5\%$), the degree of reaction for the baseline is 0.27, increases to 0.36 for the steady-optimized turbine (profile 3, Fig. 6), and increases to 0.29 for the unsteady-optimized turbine (profile 2, Fig. 8a). Table 2 lists the total pressure loss of the stator ($1 - \frac{P_{02}}{P_{01}}$) and rotor ($1 - \frac{P_{03R}}{P_{02R}}$), evaluated for selected profiles at steady and full unsteady simulations. With increasing inlet fluctuation amplitude, the vane and rotor total pressure losses increase, due to appearance of unsteady recirculation regions. The endwall optimization provides an improvement in the vane performance and the improvement in the vane is remarkably higher than in the rotor. Additionally, preliminary unsteady evaluations indicate that diffuser loss may be 30% depending on flow speed and damping. Hence, there is a need for a well-designed upstream diffuser that reduces the flow fluctuations. Fig. 10a depicts the mid-span relative Mach contour of the steady-optimized turbine (pro-

file 3, Fig. 6) during one period of the inlet fluctuations and turbine efficiencies are plotted in Fig. 10b. The steady-optimized turbine (profile 3, Fig. 6) outperforms the baseline (by 10 percentage-points) and unsteady-optimized turbine (by 0.4 percentage-point) when operating at steady state. However, under pulsating conditions, the full unsteady assessment demonstrates that the unsteady-optimized turbine (profile 2, Fig. 8a) outperforms the baseline and steady-optimized turbine (profile 3, Fig. 6) by up to 13 percentage-points. The deviation in terms of turbine efficiency between the full unsteady analysis and the mixing plane approach was 4.4 percentage-points, which demonstrates the validity of using the mixing plane approach for the unsteady optimization.

$$R = \frac{h_2 - h_3}{h_{01} - h_{03}} \quad (11)$$

3.4. Thermal powerplant analysis

To assess the benefits of rotating detonation engines with the retrofitted turbine, the engine core of several engine configurations were analyzed via an engine model with T-MATS [46]. T-MATS is a NASA developed open source toolkit to model thermodynamic systems such as engines and turbomachinery components. Three different components of the engine core were implemented. A compressor, which requires a compression ratio (Π) and efficiency (η_{is}) as input, the RDC and the turbine. The triangles in Fig. 11 depict the pressure gain ($P_{0,outlet}/P_{0,inlet} - 1$) of the rotating detonation combustor for several overall compression ratios for stoichiometric fuel-air mixture, calculated with a reduced order model [15] (verified through 2D URANS simulations). This RDC model is based on Cantera (open-source chemical kinetics library) to model the detonation properties across the shock and a method of characteristics to model the expansion region downstream of the shock. The model pro-

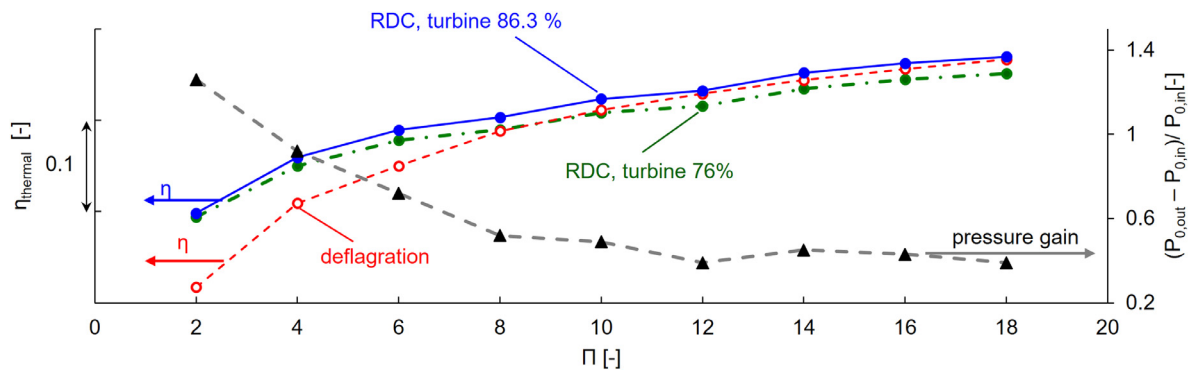


Fig. 11. Pressure gain and cycle efficiency for several engine cores.

vides the unsteady RDC outlet and the mass-flow averaged values. The diffuser, mentioned in Section 2.5, is modeled with a 30% total pressure loss. The last component is the high-pressure turbine which in contrast to supersonic turbines [1], has an outlet Mach number of 0.36.

Fig. 11 details the thermal efficiency ($\frac{(\dot{m}_0 + \dot{m}_5)V_5^2 - \dot{m}_0 V_0^2}{2\dot{m}_f Q_f}$) of the engine core for a turbojet application in function of the overall compression ratio for several engine layouts. This engine core is evaluated with two different turbines: a turbine with an efficiency of 86.3%, profile 2 in Fig. 8a with 25% mean-to-max amplitude; and a second one with an efficiency of 76%, the baseline profile with 25% mean-to-max amplitude. At low pressure ratios, benefits are almost 10 percentage-points. At intermediate compression ratios, the turbine efficiency becomes increasingly important.

Let us consider as baseline engine the Rolls-Royce M250-C20B series, a turboshaft for helicopters. The overall pressure ratio is 7.2:1 an airflow rate of 1.8 kg/s. At this pressure ratio, the pressure gain is 0.62 (Fig. 11). The required compressor pressure ratio can be reduced to 6.35 to deliver the same work to the low pressure turbine. According to (Eq. (12)) the compressor power demand can therefore be reduced by up to 8% compared to the deflagration engine implying a similar reduction in fuel consumption.

$$\text{compressor power} \sim \dot{m}_c p \frac{T_{0, \text{inlet}}}{\eta_c} \left(\Pi^{\frac{\gamma-1}{\gamma}} - 1 \right) \quad (12)$$

This lower demand in compressor power could lead to overall smaller and lighter components and additionally the efficiency of the compressor and high-pressure turbine can be improved. Finally, lower compression ratios across the compressor could also further boost pressure gains due to a decrease in total temperature at the inlet of the combustor.

4. Conclusions

This article presents a comprehensive study to upgrade current thermal power plants based on deflagration combustion to a pressure gain combustion system. Specifically, a novel design method is proposed to increase the turbine efficiency downstream of pressure gain combustors with a short diffuser. The performance benefits of adapting the current thermal power plants to a configuration based on an RDC, diffuser, and optimized turbine are demonstrated through an engine model of the upgraded power plant, which yielded an increase of thermal efficiency almost 10 percentage-points at low pressure ratios and power reduction of the compressor for a turboshaft by 8%.

The baseline turbine exhibits flow separation near the endwalls, hence the turbine endwalls should be optimized while ensuring that the turbine passage is started. Inspired by this challenge, we propose a multi-step optimization approach that allows the retrofit of a turbine, without altering the shape of the turbine airfoil and only changing the end-wall profile, with the turbine efficiency as primary objective func-

tion. In a first step, the baseline profile is optimized for a steady inlet using 3D RANS simulations. An efficiency increase of 12.5% is obtained with reduced flow separation and total pressure loss, compared to the baseline geometry. The critical design parameters are identified and fed to the second optimization step in which the turbine inlet is submitted to large amplitude fluctuations from a diffuser. The endwall profiles are further optimized by solving the 3D URANS equations and a mixing plane method. An efficiency increase of 21.4% is achieved compared to the baseline. Finally, full unsteady simulations, which account for the fully resolved stator-rotor interactions, are carried out on selected optimized profiles and demonstrated that the unsteady-optimized profile outperforms the baseline design by up to 13 percentage-points. In contrast to the baseline turbine, where the stator passage acts as a diverging-converging channel, the diffusion is initiated upstream of the stator for the optimized profiles resulting in a converging channel along the stator passage so that flow turning can be maximized without separating the flow.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Zhe Liu: Formal analysis, Investigation, Writing - original draft.
James Braun: Methodology, Investigation, Writing - original draft.
Guillermo Paniagua: Supervision, Conceptualization, Methodology, Writing - original draft.

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Table 3
Sutherland law parameters for viscosity and thermal conductivity of air.

	T_{ref}	$\sigma_{\text{viscosity,ref}}$	$C_{\text{viscosity}}^{\text{te}}$	k_{ref}	$C_{\text{thermal conductivity}}^{\text{te}}$
Air	273.11	1.716e-5	111.0	0.0241	194.0

Annex. Governing equations, constitutive equations and gas thermodynamic properties

$$\frac{\partial Q}{\partial t} + \frac{\partial(F_1 + G_1)}{\partial x} + \frac{\partial(F_2 + G_2)}{\partial y} + \frac{\partial(F_3 + G_3)}{\partial z} = \dot{S} \quad (13)$$

Q describes the dependent variable vector, F the inviscid flux vectors, and G the viscous flux vector:

$$Q = \begin{pmatrix} e \\ \rho \\ \rho u \\ \rho v \\ \rho w \end{pmatrix} F1 = \begin{pmatrix} u(e+p) \\ \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho wu \end{pmatrix} F2 = \begin{pmatrix} v(e+p) \\ \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \end{pmatrix} F3 = \begin{pmatrix} w(e+p) \\ \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \end{pmatrix} \quad (14)$$

$$G1 = \begin{pmatrix} \dot{q}_x - u\tau_{xx} - v\tau_{xy} - w\tau_{xz} \\ 0 \\ -\tau_{xx} \\ -\tau_{xy} \\ -\tau_{xz} \end{pmatrix} G2 = \begin{pmatrix} \dot{q}_y - u\tau_{yz} - v\tau_{yy} - w\tau_{yz} \\ 0 \\ -\tau_{yx} \\ -\tau_{yy} \\ -\tau_{yz} \end{pmatrix} G3 = \begin{pmatrix} \dot{q}_z - u\tau_{zx} - v\tau_{zy} - w\tau_{zz} \\ 0 \\ -\tau_{zx} \\ -\tau_{zy} \\ -\tau_{zz} \end{pmatrix} \quad (15)$$

τ represents the shear stress vector, q the heat flux vector and $\bar{k}_m$ the thermal conductivity:

$$\begin{aligned} t_{xx} &= 2\bar{\mu}_m \frac{\partial u}{\partial x} - \frac{2}{3}\bar{\mu}_m \theta, t_{yy} = 2\bar{\mu}_m \frac{\partial v}{\partial y} - \frac{2}{3}\bar{\mu}_m \theta, t_{zz} = 2\bar{\mu}_m \frac{\partial w}{\partial z} - \frac{2}{3}\bar{\mu}_m \theta, t_{xy} = t_{yx} \\ &= \bar{\mu}_m \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ t_{xz} &= t_{zx} = \bar{\mu}_m \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right), t_{yz} = t_{zy} = \bar{\mu}_m \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right), \theta \\ &= \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right), \dot{q}_x = -\bar{k}_m \frac{\partial T}{\partial x} \\ \dot{q}_y &= -\bar{k}_m \frac{\partial T}{\partial y}, \dot{q}_z = -\bar{k}_m \frac{\partial T}{\partial z} \end{aligned} \quad (16)$$

The thermodynamic properties were evaluated based on a real gas model with a temperature-dependent coefficient polynomial for working fluid with the coefficients found in Bride et al. [47]. The Sutherland Law was used for the viscosity calculations. Table 3 lists the values of temperature-dependent parameters for air.

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